

NETSO ENERGY

Rooftop Solar RESCO / OPEX Platform

RESIDENTIAL BUILDING — COMPLETE FINANCIAL MODEL

6-Storey Building | Chittagong | BDT 40,000–60,000/month Bill

Scenario A: Solar Only vs Scenario B: Solar + HiTHIUM Battery

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EXECUTIVE SUMMARY

Netso Energy is a Rooftop Solar RESCO (Renewable Energy Service Company) operating on an OPEX model. Netso finances, owns, and operates rooftop solar systems on customer buildings. Customers pay a below-grid per-unit tariff with zero upfront cost. Netso generates recurring monthly revenue over a 10–15 year contract term.

What This Report Covers

This model analyses a single deployment on a 6-storey, 12-apartment residential building in Chittagong with a monthly electricity bill of BDT 40,000–60,000. Two scenarios are stress-tested: solar only (Scenario A) and solar plus HiTHIUM MaxPower 16 battery storage (Scenario B).

Recommended System Size

22 kW

Netso Monthly Profit (Loan)

BDT 9,502

Monthly Revenue to Netso

Netso Monthly Profit (Yr 11+)

BDT 22,638**BDT 21,721**

Customer Monthly Saving

BDT 7,350+

Your Equity Required

BDT 275,132**Bottom Line**

The model works. Customer saves BDT 7,350/month from day one with zero upfront cost. Netso earns BDT 9,502/month net profit per building during the IDCOL loan period. After loan repayment in year 10, monthly profit jumps to BDT 21,721 — nearly 2.3x — creating the primary wealth-building event.

1. BUILDING BASELINE & CONSUMPTION

Before sizing the system, we establish the building's consumption profile by reverse-engineering from the monthly bill.

1.1 Building Profile

Parameter	Value
Building Type	6-storey residential
Number of Apartments	~12 units
Location	Chittagong, Bangladesh
Monthly Electricity Bill	BDT 40,000 – 60,000 (modelled at BDT 50,000)
BPDB Tariff Category	LT-A Residential
Estimated Total Consumption	5,000 kWh/month
Avg. Effective Rate (incl. VAT)	~BDT 10.00/kWh blended

1.2 BERC Official Tariff Slabs (LT-A Residential)

The following are the official BERC-approved residential tariff slabs. A 12-apartment building with high consumption regularly falls into the upper slabs, making the solar savings case very compelling.

Consumption Slab	Rate (BDT/kWh)	Relevance to This Building
0 – 75 units	5.26	Minimal — covered in first hours of day
76 – 200 units	7.20	Partial
201 – 300 units	7.59	Partial
301 – 400 units	8.02	Active
401 – 600 units	12.67	★ High — building is here most months
601+ units	14.61	★★ Peak — this is your pricing headroom
Demand Charge	BDT 42/kVA	Fixed component

Key Insight: Pricing Headroom

The building's marginal grid rate is BDT 12.67–14.61/kWh. Netso charges BDT 7.00/kWh — a 45–52% discount. This is your sales pitch. The savings case is immediate and undeniable.

2. SOLAR RESOURCE — CHITTAGONG

Season	Peak Sun Hours/Day	Monthly kWh per kWp
Nov – Feb (Winter)	5.8 – 6.2 hrs	174 – 186 kWh/kWp
Mar – May (Pre-monsoon)	5.0 – 5.5 hrs	150 – 165 kWh/kWp
Jun – Sep (Monsoon)	3.2 – 3.8 hrs	96 – 114 kWh/kWp
Oct (Post-monsoon)	4.5 – 5.0 hrs	135 – 150 kWh/kWp
Annual Average	4.9 hrs/day	~147 kWh/kWp/month

System Sizing Logic

We size to maximise self-consumption, not total generation. Oversizing causes excessive export at low BPDB export rates (BDT 3.50–5.00/kWh), reducing ROI. Target: 75–85% of generation consumed directly. For 5,000 kWh/month consumption with ~50–60% occurring in daytime hours, the optimal system is 22 kW.

3. SYSTEM DESIGN & CAPITAL COST

3.1 Recommended System: 22 kW

Sizing Scenario	Monthly Generation	Self-Consumed
Conservative — 15 kW (30% offset)	2,205 kWh	1,985 kWh (90%)
★ OPTIMAL — 22 kW (45% offset)	3,234 kWh	2,587 kWh (80%)
Aggressive — 30 kW (60% offset)	4,410 kWh	3,087 kWh (70%)

3.2 Capital Expenditure Breakdown

Component	Scenario A (Solar Only)	Scenario B (+ Battery)
Solar Panels (22 kW @ BDT 43.5/W)	BDT 957,000	BDT 957,000
Inverter + Balance of System (20%)	BDT 191,400	BDT 191,400
Aluminium Racking + Mounting (10%)	BDT 95,700	BDT 95,700
Wiring + Installation Labour (8%)	BDT 76,560	BDT 76,560
Generation Meter + Net Meter	BDT 25,000	BDT 25,000
Commissioning + Miscellaneous	BDT 30,000	BDT 30,000
HiTHIUM MaxPower 16 (16 kWh, landed)	—	BDT 260,000
TOTAL CAPEX	BDT 1,375,660	BDT 1,635,660

3.3 IDCOL Financing Structure

Financing Item	Scenario A	Scenario B
CAPEX	BDT 1,375,660	BDT 1,635,660
IDCOL Loan (80%)	BDT 1,100,528	BDT 1,308,528
Your Equity Required (20%)	BDT 275,132	BDT 327,132
Interest Rate	6% per annum	6% per annum
Loan Tenure	10 years	10 years
Monthly EMI	BDT 12,219	BDT 14,527

4. REVENUE MODEL & PRICING STRATEGY

Netso's revenue model is straightforward: charge the customer a fixed per-kWh rate on total system generation, metered at source. The customer keeps all net metering export credits from BPDB. This creates a clean, auditable, dispute-free billing relationship.

4.1 Pricing Rationale

Rate Comparison	Per kWh (BDT)	Context
Customer's current marginal grid rate	12.67 – 14.61	Upper slabs (LT-A)
Netso OPEX charge to customer	7.00	Fixed for contract term
Customer saving per kWh (solar units)	5.67 – 7.61	45–52% discount
BPDB net metering export rate	3.50 – 5.00	Kept by customer (bonus)
Netso LCOE (validated)	4.91 – 5.50	Cost per kWh generated
Netso gross margin per kWh	1.50 – 2.09	Spread over LCOE

4.2 Contract Protection Clause

Minimum Generation Payment Clause (Essential)

The OPEX contract must include: Customer pays for a minimum of 85% of the system's rated monthly generation (2,749 kWh/month) regardless of actual consumption. This protects Netso's revenue floor at BDT 19,243/month minimum even if the building reduces consumption or installs energy-efficient appliances.

5. MONTHLY P&L — BOTH SCENARIOS

5.1 Generation & Consumption Flow

Energy Flow Item	Scenario A (Solar Only)	Scenario B (+ Battery)
Total Solar Generation/month	3,234 kWh	3,234 kWh
Directly Self-Consumed (Daytime)	2,587 kWh (80%)	2,340 kWh (72%)

Energy Flow Item	Scenario A (Solar Only)	Scenario B (+ Battery)
Stored in Battery	—	390 kWh
Battery Discharged at Night	—	370 kWh (95% efficiency)
Total Self-Consumed (Direct + Battery)	2,587 kWh	2,710 kWh
Exported to Grid (Net Metered)	647 kWh	524 kWh
Remaining Grid Consumption	2,413 kWh	2,290 kWh
Grid Independence Achieved	51.7%	54.2%

5.2 Customer Bill: Before vs. After

Bill Component	Before Netso	Scenario A	Scenario B
Pays Netso (solar generation)	—	BDT 22,638	BDT 22,638
Pays BPDB (remaining grid units)	BDT 50,000	BDT 22,924	BDT 21,755
Net Metering Credit from BPDB	—	(BDT 2,912)	(BDT 2,358)
TOTAL MONTHLY COST	BDT 50,000	BDT 42,650	BDT 42,035
Monthly Saving	—	BDT 7,350	BDT 7,965
Saving %	—	14.7%	15.9%

5.3 Netso Monthly Profit & Loss

P&L Item	Scenario A (Solar Only)	Scenario B (+ Battery)
Revenue from Customer	BDT 22,638	BDT 22,638
IDCOL EMI Repayment	(BDT 12,219)	(BDT 14,527)
O&M Cost (0.5% CAPEX/yr ÷ 12)	(BDT 573)	(BDT 681)
Insurance (0.3% CAPEX/yr ÷ 12)	(BDT 344)	(BDT 409)
NET MONTHLY PROFIT	BDT 9,502	BDT 7,021
Annual Profit	BDT 114,024	BDT 84,252
10-Year Cumulative Profit	BDT 1,140,240	BDT 842,520

6. POST-LOAN WINDFALL — YEAR 11 ONWARD

This is the most important section in the entire model. Once the IDCOL loan is repaid after year 10, the monthly cost structure collapses to O&M and insurance only. Revenue continues unchanged. Profit nearly triples.

Phase	Scenario A	Scenario B
Monthly Revenue (unchanged)	BDT 22,638	BDT 22,638
Monthly Costs (yr 11+: O&M + Insurance only)	BDT 917	BDT 1,090
MONTHLY PROFIT (YR 11+)	BDT 21,721	BDT 21,548
ANNUAL PROFIT (YR 11+)	BDT 260,652	BDT 258,576
Increase vs. Loan Period	+128.6%	+207.0%

The Wealth-Building Event

Year 10 is the inflection point. IDCOL is repaid. The system is fully paid off. For the remaining 5–10 years of the contract, Netso collects BDT 21,721/month per building in near-passive profit. Across 5 buildings, this is BDT 108,605/month (BDT 1.3M/year) in recurring income with minimal overhead. This is the real reason to deploy fast and build the portfolio now.

7. HEAD-TO-HEAD SCENARIO COMPARISON

Metric	Scenario A: Solar Only ★	Scenario B: + HiTHIUM Battery
System CAPEX	BDT 1,375,660	BDT 1,635,660
Your Equity Required	BDT 275,132	BDT 327,132
Monthly Revenue	BDT 22,638	BDT 22,638
Monthly Profit (Loan Period)	BDT 9,502 ★	BDT 7,021
Monthly Profit (Yr 11+)	BDT 21,721	BDT 21,548
Annual Profit (Loan Period)	BDT 114,024 ★	BDT 84,252
Customer Monthly Saving	BDT 7,350	BDT 7,965 ★
Grid Independence	51.7%	54.2% ★

Metric	Scenario A: Solar Only ★	Scenario B: + HiTHIUM Battery
Equity Payback Period	28 months ★	46 months
10-Yr Cumulative Profit	BDT 1,140,240 ★	BDT 842,520
Installation Complexity	Low ★	Medium
RECOMMENDATION	DEPLOY NOW (Deals 1–3)	Deploy from Deal 4+

8. PORTFOLIO SCALING — WHAT 5 BUILDINGS LOOKS LIKE

The OPEX model is designed to be replicated rapidly. Each building is an independent IDCOL-financed project. The more buildings deployed, the stronger the portfolio cash flow — even while individual loans are still being repaid.

Metric	1 Building	2 Buildings	3 Buildings	5 Buildings
Monthly Revenue	22,638	45,276	67,914	113,190
Monthly Profit (Loan)	9,502	19,004	28,506	47,510
Monthly Profit (Yr 11+)	21,721	43,442	65,163	108,605
Annual Revenue	271,656	543,312	814,968	1,358,280
Equity Needed (Total)	275,132	550,264	825,396	1,375,660
Customers Benefited	12 apts	24 apts	36 apts	60 apts

Path to BDT 1M/Year Revenue

4 buildings at scale = BDT 1.08M annual revenue during loan period. 5 buildings post-loan = BDT 1.3M/year near-passive. Each building requires BDT 275,132 in equity — totalling BDT 1.375M for 5 buildings. The IDCOL loan does the heavy lifting on each project.

9. STRATEGIC RECOMMENDATIONS

9.1 Immediate Actions (This Week)

- Call every warm residential contact. Confirm 3 site visits.
- Visit each rooftop: photograph, measure, collect 3 months of electricity bills.
- Request 3 hardware quotes from Dhaka suppliers for a 22 kW system.
- Prepare a 1-page proposal per building: current bill vs. Netso OPEX charge.
- Target one signed LOI within 14 days.

9.2 Pricing & Contract Rules

- Always charge BDT 7.00/kWh — never less. You have massive headroom vs. the BDT 12.67–14.61 marginal grid rate.
- Include minimum generation clause: customer pays for 85% of rated monthly generation regardless of consumption.
- Give the customer all net metering credits — it costs you nothing and accelerates deal closure.
- Contract term: 12 years minimum. System transfer to customer at end of term.

9.3 Battery Strategy

- Deals 1–3: Solar only. Higher margin (BDT 9,502/month), faster equity payback (28 months), simpler installation.
- Deal 4 onward: Add HiTHIUM MaxPower 16 per building as cash flow from earlier deals funds part of the battery cost.
- Request a formal quote from HiTHIUM now (Bangladesh is listed as a supported region on their website).
- Sodium-ion batteries: revisit in 2027–2028 when CATL/BYD products reach Bangladesh distributors at scale.

9.4 Financing Path

- IDCOL primary: 80% debt at 6% over 10 years. Submit pre-qualification immediately with your SREDA registration.
- Parallel: approach BRAC Bank and Dutch-Bangla Bank SME green energy products as backup.
- Supplier credit: negotiate 30–50% upfront, balance on commissioning with your hardware supplier.
- Your equity per building: BDT 275,132 (~\$2,300 USD). Each deal is self-contained.

10. KEY RISKS & MITIGATIONS

Risk	Impact	Mitigation
Customer reduces consumption over time	Your revenue/kWh stays same but their savings shrink — deal looks worse	Minimum generation payment clause in contract

Risk	Impact	Mitigation
BPDB tariff decreases	Your BDT 7.00/kWh offer less attractive	Lock tariff discount % not absolute rate in contract
IDCOL application delayed	Capital stuck, deal stalls	Parallel bank application from day 1
Net metering approval delay	Customer can't export surplus for months	Start with common-area-only system to avoid NEM dependency
Panel degradation over 10 years	~0.5%/yr, reduces generation by ~5% by year 10	Built into yield estimates; minor impact on revenue
20-year contract enforcement in Bangladesh	High legal risk if customer defaults or disputes	Shorter 12-year term, strong LOI, physical ownership of system as collateral

FINAL VERDICT

Monthly Revenue (per building)

BDT 22,638

Customer Saves Per Month

BDT 7,350+

Monthly Profit During Loan

BDT 9,502

Your Equity Per Building

BDT 275,132

Monthly Profit After Loan (Yr 11)

BDT 21,721

Equity Payback Period

28 months

The model is viable. The deal sells itself.

A 6-storey Chittagong building owner pays BDT 50,000/month today. With Netso, they pay BDT 42,650. Zero upfront. System maintained by Netso. After 12 years the system is theirs for free.

Netso earns BDT 9,502/month per building during the loan period — and BDT 21,721/month after the loan is repaid.

The only remaining question is: who are you calling first?